

UNIT – III

Biodiversity and Biotic Resources

The term species refers to all organisms of the same kind that are genetically similar enough to breed in nature and produce live and fertile organisms.

Diversity is a combination of a number of species and the number of individuals of each species in a community.

On earth-	1 million	- animal species
	4000species	- mammals
	10,000 species	- birds
	6,000 species	- reptiles
	2,500 species	- fishes
	4.5 million	- invertebrates
	0.5 million	- plants.

3.1. Assessment of biodiversity:

Bio diversity is analyzed or assessed generally at the following four levels:

1. Diversity of composition of a species within a community:

The biotic components of an ecosystem may be composed of a number of species of plants, microbes and animals which interact with each other on one hand and interact with the abiotic factors. The richness of species in an ecosystem is called 'species diversity'.

2. Diversity of genetic organization within a species:

Within any given species, we can find several variations which slightly differ from each other in one or more characteristics such as size, shape, quality of their respective product, resistance against pests, insects, diseases etc.

Such diversity in the genetic make-up of a species is called 'genetic diversity'.

3. Diversity of Biotic communities and ecosystems:

An ecosystem develops its own characteristic community of living organisms depending upon the availability of abiotic resources, environmental conditions and other relevant factors.

Different types of ponds, lakes, rivers, wetlands, grasslands, forests etc. represent diverse ecosystems with their own characteristic biotic community.

For example in forest ecosystem biotic components are different compared to another ecosystem such as grassland.

	Ecosystems	Producers	Consumers
1.	Forest	Trees	Rabbit, Monkey, Lion, Tiger, Deer etc.
2.	Desert	desert plants Ex. Opuntia, Cactus	Camel, Snake, lizards etc.
3.	Pond	Macro and Micro- Phytes ex: algae, lotus	Zooplanktons, water beetles, fish etc

Diversity in biotic communities

4. Diversity within a land- scape: This refers to the size and distribution of several ecosystems and their interactions across a given land surface.

3.2.Value of Biodiversity:

Biosphere is a life supporting system to the human beings. It is the combination of different organisms. Each organisms in the biosphere has its own significance. Biodiversity is vital for healthy biosphere. Biodiversity is must for the stability and proper functioning of the biosphere.

We get benefits from other organisms in number of ways. Some times we realize the real value of the organism only after it is lost in this earth.

The values of biodiversity have been classified as follows:

1. Direct values: These are assigned to the products harvested by people, Direct values can be further divided as follows:

a. Consumptive use value: This can be assigned to goods such as fuel wood and gum that are consumed locally and do not figure in national and international market.

Examples:

- i. Food: A large number of wild plants consumed by human beings as food. Nearly 80-90% of our food crops have been domesticated only from the tropical wild plants. A large number of wild animals are also consumed as food.
- ii. Drugs: Around 70% of modern medicines are derived from plant extracts. 20,000 plant species are believed to be used medicinally, particularly in the tradition system of Unani, Ayurveda and Sidha.

Ex:

- India uses 3000 species of plants in Ayurveda, Homeopathy and Unani system of medicines.
- Life saving drug like Quinine (Malaria), Reserpine (Hypertension), Pencillin (Antibioti) and Morphine (Painkiller) are all of plant origin.
- The Peepal tree leaves, trunk and roots are used as effective medicines for curing diseases like fever, cough, stomach and skin diseases.

- About 30 medicines have been prepared from Neem tree which have been proved to be very effective for stomach ailments, eye irritations, skin eruptions and diabetics.

Medicinal products from Natural resources

Products	Source	Use
Penicillin	Fungus	Antibiotic
Streptomycin	Actinomycete	Antibiotic
Tetracycline	Bacterium	Antibiotic
Digitalis	Foxglove	Heart stimulant
Quinine	Cinchona Bark	Malaria treatment
Diosgenin	Yarn	Birth control drug
Cytarabine	Sponge	Leukemia cure
Reserpine	Rauwolfia	Hypertension drug
Bee venom	Bee	Arthritis relief
Morphine	Poppy	Analgesic

iii. Fuel: Fire woods are directly consumed by villages, tribal. The fossil fuels like coal, petroleum and natural gas are also the products of fossilized biodiversity.

b. Productive use value: This is assigned to products that are harvested from the wild and sold in commercial markets, both at national and international levels.

Animal Products

Animal product	Animal
Silk	Silk-worm
Wool	Sheep
Musk	Musk deer
Tusk	Elephants
Leather	All animals
Food	Fish and animals

Many industries are dependent upon the productive use values of biodiversity.

Product	Industry
Wood	Paper and pulp, Plywood, railway sleeper industries.
Cotton	Textile industry
Fruits, vegetables	Food industry
Leather	Leather industry
Ivory	Ivory – works
Pearl	Pearl industry

2. Indirect values: these are assigned to benefits provided by biodiversity that do not involve harvesting or destroying the resource means without consumption of resource. Such benefits include water quality, soil protection, recreation, and education, and scientific research, regulation of climatic and producing further options for human society.

Indirect values further divided into-

a. Non consumptive use value: This includes ecosystem productivity, protection of water resources, soil protection, climate regulation, waste disposal, recreation and ecotourism, environmental monitoring etc.

b. Optional value: This value of a species provide benefits in economic and research development in the further. The growing biotechnology industry is finding new species to tackle pollution, and fight problems of cancer, AIDS etc.

c. Aesthetic Value: The beautiful nature of plants and animals insist us to protect the biodiversity. The most important aesthetic value of biodiversity is eco-tourism.

d. Ethical or existence value: It involves ethical issues like “all life must be preserved”. In India and in other countries biodiversity is considered to have great value on religious and cultural basis. Our rich heritage teaches us to worship plants, animals, rivers and mountains. The ethical value means that a species may or may not be used, but its existence in nature gives us pleasure.

e. Social Value: Social values of the biodiversity refer to the manner in which the bio-reserves are used to the society. These values are associated with the social life, religion and spiritual aspects of the people.

3.3. Biodiversity hot-spots:

The areas which support rich biodiversity, because of geologic formations and endemic flora and fauna, and exhibit exceptional scientific interest are called as hot-spots.

Criteria for recognizing hot spots:

1. The richness of the endemic species is the primary criterion for recognizing hot spots.

2. The hot spots should have a significant percentage of specialized species.
3. The site is under threat.
4. It should contain important gene pools of potentially useful plants.

Reason for rich biodiversity in the Tropics:

1. The tropics have a more stable climate.
2. Warm temperatures and high humidity in the tropical areas provide favorable conditions.
3. No single species can dominate and thus there is an opportunity for many species to co exist.
4. Among plants, rate of out –crossing appear to be higher in tropics.

At global level Myers (1988) has identified 10 hot spots of tropical forests to which he later added 8 areas. These areas collectively include 49,955 endemic species which constitute 20% of the world's total plant species. Two regions from India namely the Eastern Himalayas and the Western Ghats are included as hot spots.

Broadly the botanical hot spots lie in the i) Western Ghats ii) North-east India iii) Himalayas iv) Andaman and Nicobar Islands.

The Western Ghats particularly the Southern Western Ghats known as Malabar is the major genetic estate with an enormous biodiversity of ancient lineage.

Northwest India represents the transition zone between the Indian, Indo-Malayan, Indo-Chinese biogeographically regions as well as meeting place of Himalayan mountains with that of peninsular India.

Western Himalayas is quite remarkable for its alpine flora. Gymnosperms are quite abundant. About 5000 flowering plants occur here of which 800 species are endemic.

The Andaman and Nicobar islands have a rich flora under littoral and inland types. About 2500 species of flowering plants are found here of which 250 species are endemic.

Sacred forests are real hot spots of biodiversity. They are protected by tribal due to some religious sanctity attached to them. This forest contains no. of rare, endangered, endemic or interstices biota of the country.

These are known as 'Devaskadu' in Karnataka.

'Devarahati' in Maharashtra.

'Lakyantok' in Meghalaya.

Mangrove forests are salt tolerant forest ecosystems. Following mangrove areas must be considered as hot spots for conservation purpose- Andaman and Nicobar, East coast (Godavari and Krishna deltaic mangroves), West coast (Vembanad in Kerala).

3.4. Major threats to biodiversity:

Extinction, the elimination of a species, is a normal process in nature.

Causes of extinction:

The causes of extinction are broadly grouped into five risk categories- population risk, environmental risk, natural catastrophe, genetic risk and human actions.

1. Population risk: Random variations in population rates (i.e. birth rates and death rates) can cause a species in low abundance to become extinct. It is a risk especially to species

in low abundance to become extinct. It is a risk especially to species that consist of only a single population in one habitat. For example – blue whales. As they swim over the vast areas of ocean, and if in one year most whales were unsuccessful in finding a mate then births could be dangerously low.

2. **Environmental Risk:** Environmental risk means variation in the physical or biological environment, including variations in predators, prey, symbiotic or competitor species.

3. **Natural Catastrophe:** a natural catastrophe is a sudden change in the environment (not as a result of human action). It includes fires, storms, floods, earthquakes, volcanic catastrophe may cause the local extinction of most forms of life there.

4. **Genetic Risk:** Detrimental change in genetic characteristics in a small population of a species, due to reduced genetic variation, genetic drift or mutation, makes the species more vulnerable to extinction because it lacks the variety once present or because a mutation that leads to poor health becomes fixed in population.

5. **Human Actions:** In the 20th century, with the introduction of industrial chemicals and emissions, pesticides etc. into the environment, pollution has become an increasingly significant cause of extinction.

Human actions cause extinction of species through:

1. Habitat loss and degradation: The main causes of habitat loss are agricultural activities, harvesting or extraction (including mining, fishing, logging, etc.) and development of human settlements, industry and associated infrastructure.

2. Habitat Fragmentation: It is a process where a large, continuous area of habitat is both, reduced in area and divided into two or more fragments. Habitat fragmentation divides population into isolated groups that not only limit the potential of species for dispersal and colonization but also reduces the foraging ability of animals. These isolated, small, scattered populations are increasingly vulnerable to inbreeding depression, high infant mortality and susceptible to environmental hardships, and consequently, in the end, possible extinction.

3. Diseases: Pathogens, or disease organisms, may also be considered predators. The incidence of disease in wild species may increase due to human activities.

4. Genetic assimilation: Some rare and endangered species are threatened by genetic assimilation because they crossbreed with closely related species that are more numerous or more vigorous. For example, hatchery-raised trout introduced into lakes or streams may genetically dilute indigenous stocks.

5. Pollution: Environmental pollution is the most harmful to habitat degradation. Most common causes of which are pesticides, industrial effluents and emissions, and emission from automobiles.

6. Poaching: Poaching is another insidious threat that has emerged in recent decades as one of the primary reasons for the decline in number of species.

Poaching of wildlife for sport, making profit and for human consumption, wild species are hunted for their fur, tusks, meat, thorns. Various animals and their purpose of producing article/ use are summarized here.

S.No.	Species	Use
1.	Elephant	Ivory, ashtray
2.	Alligators	Boots for urban needs
3.	Blubber	Lubricating oils
4.	Baleen	Combs and other articles
5.	Tiger	Skin and bones
6.	Rhinos	Horns
7.	Deer	Musk, perfume

7. Introduction of exotic species: Organisms introduced into habitats where they are not native are termed as exotic. They can be thought of as biological pollutants and are considered to be among the most damaging agents of habitat alteration and degradation in the world.

8. Man wild life conflicts: Man is continuously interacting with different ecosystems for food, fuel, recreation, sports, urban development, waste disposal etc. Every activity is responsible directly or indirectly responsible for disturbing species.

Natural forests are being deforested for timber and single species trees like teak, Sal. This monoculture plantation creates imbalance ecosystem.

Man-wildlife conflicts arise, when wildlife starts causing immense damage and danger to the man. Ex:1. In Sambalpur, Orissa, 195 human were killed in the last 5 years by elephants. In retaliation, the villagers have killed 98 elephants and badly injured 30 elephants. 2. A total of 14 persons were killed during 19 attacks by the leopards in Sanjay Gandhi National Park, Mumbai.

3.5. Categories of Species:

The International Union of conservation of Nature and Natural Resources (IUCN) maintains what may be called a 'Red Database' at the world Conservation Monitoring Centre (WCMC) in which information on endangered and vulnerable species of plants and animals is kept. From time to time, this database is translated into popular form and published as 'Red Data Books'. IUCN has developed categories that can be assigned to species of plants and animals under threat based on:

- The present and past distribution of the species;
- The decline in the number of population of the species in the course of time;
- Abundance and quality of natural habitat of the species; and
- The biology and potential value of the species.

The categories are as follows:

- i) **Endangered species (E):** The species that are considered in imminent danger of extinction and whose survival is unlikely if factors causing their decline continue to operate. These are species whose numbers have been reduced to a critical level or whose habitats have been so drastically reduced that they are in immediate danger of extinction.
- ii) **Vulnerable (V):** The species that are under threat such that they may have to be classified as endangered in the near future if causal factors continue to operate. These include species whose populations have been seriously depleted and whose ultimate security is not assured, as well as those species whose populations are still abundant but are under threat throughout their range.
- iii) **Rare species ®:** These are species with small total population size in the world. In their distribution, they are usually localized within restricted habitats or geographical area or thinly scattered over an extensive range. Rare species, thus, are not at present endangered and vulnerable but are at risks.
- iv) **Threatened species (T):** The term ‘threatened’ is used in the context of conservation of the species which are in any one of the above three categories. These are species that have declined significantly in total numbers and may be on the verge of extinction in certain localities.

Indian endangered plants:

In India, nearly 450 species have been identified as endangered threatened or rare. Some such species are-

1. Anemia Tomentosa. 2. Frerea indica 3. Populus gamblei
4. Vanda coerulea 5. Santalum Album.

Ex Extinct	EW Extinct in the wild	CR Critically endangered	EN Endangered	VU Vulnerable	LR/cd Lower risk conservation dependent	LR/nt Lower risk near threatened	DD Data deficient
7	2	44	113	87	1	72	14

Indian endemic plant species:

Endemics are species that are found in a single locality/area and no where else in the world.

Group	No. of species
Pteridophyta	200
Angiosperm	4,950
Total	5,150

Indian endangered fauna:

Some of the animal species are listed below have been identified as endangered ones-

1. **Mammals:** Hoolock gibbon (ape), Nilgiri langur, Phayre’s leaf monkey, Chinese pangolin, Jackal, Red fox, Indian fox, Wild dog, Desert cat, Jungle cat, Rhinoceros, Indian wild ass, Tibetan wild ass, Andaman wild pig, Mouse deer, Black buck, Indian bison, Assam Rabbit, Flying squirrels, Marine dolphins etc.
2. **Birds:** Geese, Swans, Eagles, etc.

3. Reptiles: Turtles, Green sea turtle, Shell turtle, Indian python, Indian egg eating snake, Man eater lizards, etc.
4. Amphibians: Viviparous toad, Indian salamander, etc.
5. Invertebrates: Coconut or Robber crab, Beetles, Dragonfly etc.

Threatened Animals of India by status category

Ex Extinct	EW Extinct in the wild	CR Critically endangered	EN Endangered	VU Vulnerable	LR/cd Lower risk conservation dependent	LR/nt Lower risk near threatened	DD Data deficient
0	0	18	54	143	10	99	31

Threatened species of India by Taxonomic group

Taxonomic Group	Number of threatened species
Mammals	86
Birds	70
Reptiles	25
Amphibians	3
Fish	3
Molluscs	2
Other Invertebrates	21
Plants	244
Total	459



Carolina parakeet



Northern blue fin tuna



Giant flightless birds called Moa were overexploited

India as a Mega-diversity Nation:

- India is one of the twelve centers of origin of cultivated plants. India has five heritage sites, 13 biosphere reserves and six wetlands. Also, India has 88 national parks and 490 sanctuaries.
- Total number of plant species in India = 45000
Cultivated plants of species = 27

- Number of endemic plant species (found in specific) areas = 4000 (18% of the world). All above figures show that India has a high biodiversity.
- A part from high biodiversity in Indian plants, cultivated crops and animals there is also great diversity of domestic livestock breeds. India has 27 indigenous breeds of cattle, 40 breeds of sheep, 22 breeds of goats and 9 breeds of buffaloes. All above features of Indian diversity show that India is a Mega-diversity nation.

3.6. Biodiversity conservation:

Biodiversity is one of the important tools for sustainable development. The enormous value of biodiversity due to their commercial, genetic, aesthetic and ecological importance emphasizes the need to conserve biodiversity.

There are two types of biodiversity conservation:

1. In-situ conservation:

Here the species are protected in their own natural or a slightly modified man-made ecosystems. In-situ conservation ex. Natural monuments, natural reserves, cultural landscapes, national parks, sanctuaries etc.

Protecting the areas helps not only in conserving individual species but preserve ecosystems also. In these protected areas – tourism, explosive activities, poaching, shooting, grazing of domestic animals, cutting of trees are strictly prohibited.

@ Biosphere reserves: These reserves cover large area, more than 5000 sq.km. It is used to protect species for long time. Ex. Sunderbans in West Bengal, Nilgiri in Karnataka, Kerala, Tamilnadu, Manas in Assam etc.

@ National Parks: A national park is an area dedicated for the conservation of wildlife along with its environment. It is usually a small reserve covering an area of about 100 to 500 sq.km. Within the biosphere reserves, one or more National parks are also exist. Ex. Kaziranga in Assam, Sariska in Rajasthan etc.

@ Wildlife Sanctuaries: A wildlife sanctuary is an area, which is reserved for the conservation of animals only. At present, there are 492 wildlife sanctuaries in our country. Ex. Sultanpur Bird Sanctuary in Haryana, Wild ass Sanctuary in Gujarat etc.

2. Ex-situ conservation:

Here the species are protected outside their habitats and increase their population in a planned way. Ex. Zoo's, Botanical gardens, culture collection centers, genetic resource centers (here seeds, pollen, semen, ova, cells etc. stored in gene pools or germ plasam banks) etc.

Advantages of Ex-situ conservation: The organisms will have longer life span because of assured food, water, shelter and security. Because of special care the species endangered may survive longer. Modern facilities can provide better process of breeding.

Disadvantages: Maintenance and cost of breeding plants and animals are expensive. Species are habitual to favorable environmental conditions, they can not adapt to ever changing natural condition.

Steps for conservation of Biodiversity:

1. Biodiversity inventories and assessments – population surveys and assessments – population surveys and assessment.
2. Identifying and expanding protected areas.
3. Conserving biodiversity in seed banks and gene banks.
4. Controlling wild life trade.
5. Providing environmental education to the people.
6. Reviewing agricultural practices.
7. Controlling urbanization.
8. Geographical information system for planning and monitoring.
9. Restoration of biodiversity.
10. Population control.
11. Implementing environmental protection Act (EPA).
12. Involving more non government organizations (NGOs).

3.7. The National Biological Diversity Act, 2002

It is an Act of the Parliament of India for preservation of biological diversity in India, and provides mechanism for equitable sharing of benefits arising out use of traditional biological resources and knowledge. The Act was enacted to meet the obligations under Convention on Biological Diversity (CBD), to which India is a party.

Biodiversity and biological resource

Biodiversity has been defined under Section 2(b) of the Act as "the variability among living organisms from all sources and the ecological complexes of which they are part, and includes diversity within species or between species and of eco-systems". The Act also defines Biological resources as "plants, animals and micro-organisms or parts thereof, their genetic material and by-products (excluding value added products) with actual or potential use or value, but does not include human genetic material".

National Biodiversity Authority and State Biodiversity Boards

The National Biodiversity Authority (NBA) is a statutory autonomous body, headquartered in Chennai, under the Ministry of Environment and Forests, Government of India established in 2003 to implement the provisions under the Act. State Biodiversity Boards (SBB) has been created in 28 States along with 31,574 Biological management committees (for each local body) across India.

Functions

1. Regulation of acts prohibited under the Act
2. Advise the Government on conservation of biodiversity
3. Advise the Government on selection of biological heritage sites
4. Take appropriate steps to oppose grant of intellectual property rights in foreign countries, arising from the use of biological resources or associated traditional knowledge.

Regulations

A foreigner, non-resident Indian as defined in clause (30) of section 2 of The Income-tax Act, 1961 or a foreign company or body corporate need to take permission from the NBA before obtaining any biological resources or associated knowledge from India for research, survey, commercial utilization. Indian citizens or body corporates need to take permission from the concerned State Biodiversity Board.

Result of research using biological resources from India cannot be transferred to a non-citizen or a foreign company without the permission of NBA. However, no such permission is needed for publication of the research in a journal or seminar, or in case of a collaborative research made by institutions approved by Central Government.

No person should apply for patent or other form of intellectual property protection based on the research arising out of biological resources without the permission of the NBA. The NBA while granting such permission may make an order for benefit sharing or royalty based on utilization of such protection.

Benefit sharing

Benefit sharing out of usage of biological resources can be done in following manner:

1. Joint ownership of intellectual property rights
2. transfer of technology
3. location of production, research development units in the area of source
4. payment of monetary and non-monetary compensation
5. setting up of venture capital fund for aiding the cause of benefit claimers

Penalties

If a person, violates the regulatory provisions he will be "punishable with imprisonment for a term which may extend to five years, or with fine which may extend to ten lakh rupees and where the damage caused exceeds ten lakh rupees such fine may commensurate with the damage caused, or with both."

Any offence under this Act is non-bailable and cognizable.

3.8. Field visit

Objective

Students will share their Zoo experience through oral and written presentations.

Materials

Camera, Zoo guides download from the web site, A4 size papers.

Preparation

In preparation for the activity, the students should divide into groups and the students should take photos of different animals during field trip and the students should make

some observations and note down. And observe how the Zoo park authorities conserving the animals. Observe the health condition of animals, nativity of the animals. Ask the Zoo people common diseases affects animals.

After the field visit the students should summarize data and compare the animals and observation with those of another group. The students may also choose to just highlight what they saw at the zoo or what surprised them about the visit. The students will articulate their thoughts through mixed media presentations using maps and handouts from the Zoo's website etc. After the presentation the students should discuss the experience to highlight the conservation measures taking in the zoo park.

Evaluation

Students may receive basic credit for completing a presentation and conducting a oral debate.

Review Questions:

1. Define biodiversity and explain different types of biodiversity.
- 2.. Write a note on Value of biodiversity.
3. Justify India as a mega diversity nation.
- 4.. Explain the various threats to and measures recommended for conservation of biodiversity.
5. Define endemic species and give advantages of conservation of biodiversity.
6. Write a note timber and non-timber forest products
7. Explain In-situ and Ex-situ conservations and suggest suitable steps for conservation of biodiversity.
8. Write detailed note on biodiversity hot-spots.
9. Discuss about National Biodiversity Act.
10. Write a note on observations during field visit.

Select the appropriate answer

1. Which of the following is not a biogeographic habitat of India as per classification [c]
a) Himalayan b) Western ghats c) Sunderbans d) Desert
2. Vinblastin and Vincristine, two anticancer drugs have been obtained from [a]
a) Periwinkle b) Cincona c) Bacterium d) Jelly fish
3. Western ghats are very rich in endemic species of [c]
a) Birds b) Lion c) Amphibians d) Turtles.
4. Which of the following hotspots of biodiversity has the maximum number of plant and vertebrate species [d]
a) Caribbean b) Tropical Andes c) Madagascar d) Indo-Burma Eastern Himalayas.
5. Which of the following is an extinct species [c]
a) Dugong b) Great Indian bustard c) Dodo d) Red panda.

6. Which of the following is an example of ex-situ conservation? [b]
a) Biosphere reserve b) Gene bank c) Sanctuary d) all of these
7. Kaziranga National park is famous for [a]
a) One-horned rhino b) Hangul c) Tiger d) Elephant
8. There are only two sanctuaries in India dealing with preservation of plants. The plants Are [b]
a) Cinchona – Orchid b) Citrus – Pitcher plant
c) Mango – Citrus d) Mango – Pitcher plant
9. Cryo preservation of plant seeds and pollen is done at a very low temperature of – [c]
196°C by using
a) Ice b) carbon tetrachloride c) Liquid nitrogen d) Ammonia
10. Which one of the following National Parks do not have tigers as their main wildlife? [a]
a) Gir b) Corbett c) Dudwa d) Ranthambore.
11. The dramatic increases in agricultural production that have been made possible by high yield "miracle crops" are called the _____. [a]
a) green revolution b) biotechnical revolution
c) bio economic revolution d) greenhouse effect
12. What is the primary reason for targeting 'biodiversity hotspots' for conservation?[d]
a) They are the only areas where species are seriously threatened in the world
b) The number of species threatened far exceeds our capacity to protect them and we can therefore only concentrate on areas of highest species diversity
c) To protect all areas of threatened species would not allow for new species to develop
d) They are areas where people do not live and conservation would therefore not be effecting the economic development of the area
13. Organization responsible for maintaining Red Data Book/Red List is [a]
(a) IUCN (b) CITES (c) WWF (d) IBWL
14. In your opinion, which is the most effective way to conserve the plant diversity of an area ? [a]
(a) By creating biosphere reserve (b) By creating botanical gardens
(c) By developing seed bank (d) By tissue culture method
15. Which one of the following is a pair of endangered species ? [d]
(a) Garden lizard and Maxican poppy (b) Rhesus monkey and Sal tree
(c) Indian peacock and Carrot grass (d) Hornbill and Indian Aconite
16. If the Bengal tiger becomes extinct [c]

- (a) Hyaenas and Wolves will become scarce
 (b) The wild areas will be safe for man and domestic animals
 (c) Its gene pool will be lost for ever
 (d) The population of beautiful animals like deer's will get stabilized.
17. Which of the following is not done in a wildlife sanctuary ? [b]
 (a) fauna is conserved (b) flora is conserved
 (c) soil and flora is utilized (d) hunting is prohibited
18. Which one of the following pairs of geographical areas show maximum biodiversity in our country ? [c]
 (a) Sunderbans and Rann of Kutch (b) Eastern ghats and West Bengal
 (c) Eastern himalaya and Western ghat (d) Kerala and Punjab
19. Genetic diversity in agricultural crops is threatened by [a]
 (a) introduction of high yielding varieties (b) intensive use of fertilizers
 (c) extensive intercropping (d) intensive use of bio pesticides
20. Which of the following is considered a hot-spot of biodiversity in India ? [b]
 (a) Aravali hills (b) Western Ghats
 (c) Indo-gangetic Plain (d) Eastern Ghats
21. Which one of the following is not included under *in-situ* conservation ? [c]
 (a) national park (b) sanctuary
 (c) botanical garden (d) biosphere reserve

Fill in the blanks

- When variations occur within a species due to new combinations of genes, this is Called genetic diversity.
- Shannon- Wiener index gives a measure of species diversity.
- Drugs, fuel wood and food derived from biodiversity represent consumptive use value of Biodiversity.
- Quinine is obtained from the bark of Cincona tree.
- In terms of plant richness, the rank of India in the world is 10th
- There are 25 biodiversity hot spots in the world, of which 2 exist in India.
- Number of species found in a small homogeneous area is known as species richness.
- Species restricted only to a particular area are called endemic
- Loss of habitat in installments leading to small scattered patches is known as habitat fragmentation
- Illegal killing of prohibited endangered animals is called Poaching
- Red data book giving the list of endangered species of plants and animals is published By IUCN (International Union for conservation of Nature and Natural resources)
- Nanda Devi, Manas and Sunderbans are examples of Biosphere reserves

True or False

1. Biodiversity not only includes the variability of all types of living organisms but also the variations within the ecosystems. **True**
2. The commercially usable value of biodiversity where the product is marketed and sold is called consumptive use value. **True**
3. When ecological surveys are carried out during a tour, it is known as ecotourism. **False**
4. Prevention of soil erosion and floods, cycling of nutrients, regulation of water cycle and reduction of global warming by the trees are examples of ecosystem service value. **True**
5. The commercial value of a tree timber is much more than its ecological services value. **False**
6. Maximum global biodiversity is much more than marine diversity. **True**
7. Amongst the animals, highest number of known living species are that of mammals. **True**
8. Terrestrial diversity is much more than marine diversity. **True**
9. India has on record 47,000 species of plants and 81,000 species of animals. **True**
10. Highest number of known living species of plant kingdom in India belong to fungi. **True**
11. Through India is one of the 12 mega diversity countries in the world, yet it is not the centre of origin of any crop species. **False**
12. As we move across large landscape gradients the gamma richness increases. **True**
13. Human encroachment into forest areas is one of the main reasons for attack by wildlife on humans. **True**
14. Endangered animals cannot be prevented from becoming extinct by providing them special protection. **False**